

Gireesh Naidu Adireddi

AI/ML Engineer

aadireddigireesh2003@gmail.com · +91-9014139609 · Vizianagaram, AP, India · [LinkedIn](#) · [GitHub](#)

Professional Summary

Self-taught AI/ML Engineer with hands-on experience building, evaluating, and deploying end-to-end systems across classical ML, deep learning, NLP, and LLM-based applications. Independently designed and shipped multiple production-grade projects – including transfer learning-based computer vision models and framework-free RAG pipelines – gaining direct, practical understanding of every component. Comfortable across the full lifecycle, from development to live deployment. Seeking an entry-level AI/ML Engineer role.

Technical Skills

ML & Algorithms:	Linear Regression, Logistic Regression, Decision Trees, Random Forest, Gradient Boosting, XGBoost, SVM, KNN, Naive Bayes, Neural Networks, CNN, Transfer Learning (MobileNetV2), Backpropagation, Gradient Descent
NLP & LLMs:	Tokenization, TF-IDF, Word2Vec, Cosine Similarity, spaCy, NLTK, Transformer Architecture (Attention, Encoder-Decoder), Hugging Face Transformers, Prompt Engineering, LLM APIs, RAG Pipelines, Vector Databases (ChromaDB), Sentence Embeddings
ML Engineering:	Feature Engineering, EDA, Cross-Validation, GridSearchCV, Data Augmentation, PCA, K-Means Clustering, One Hot Encoding, Outlier Detection, Pipeline Design
SQL & Databases:	Joins, CTEs, Window Functions, Subqueries, Aggregations, Stored Procedures, Triggers, Indexing, Query Optimization, T-SQL, SSMS, Relational Data Modelling
Libraries & Tools:	Python, Scikit-learn, XGBoost, TensorFlow, Keras, OpenCV, Streamlit, NumPy, Pandas, Matplotlib, Seaborn
Deployment & Dev:	Hugging Face Spaces, Render, Streamlit Cloud, Git, GitHub, Google Colab, VS Code

Projects

Handwritten Equation Solver

[GitHub](#) | [Live App](#)

Python, TensorFlow, Keras, OpenCV, Streamlit, Docker

- Built a real-time CNN-based application that segments and recognizes handwritten math equations drawn on a canvas, achieving 95.89% validation accuracy on 8,384 images across 15 symbol classes.
- Engineered an OpenCV contour detection and smart merging pipeline to isolate individual symbols from freehand drawings, including multi-part symbols such as the division sign.
- Applied Data Augmentation (rotation, shift, zoom, shear) to improve generalization across diverse user drawing styles; deployed live on Hugging Face Spaces via Docker.

Zoro – E-commerce Customer Assistant (v1 → v2 → v3)

[GitHub](#) | [Live App \(v3\)](#)

Python, Scikit-learn, NLP, LLM APIs, RAG, ChromaDB, Groq, Streamlit

- **v1 (Classical NLP):** Built a dual-pipeline chatbot combining TF-IDF + Multinomial Naive Bayes (96% accuracy on 6,000+ records) with TF-IDF + Cosine Similarity FAQ matching; documented real synonym-gap and keyword-collision limitations through systematic edge-case testing.
- **v2 (LLM-Powered):** Rebuilt using Groq API + prompt engineering (system prompts, scope restriction, anti-hallucination guardrails), solving every documented v1 failure; discovered and fixed genuine hallucination and scope-drift bugs during testing.
- **v3 (RAG-Powered):** Implemented a full RAG pipeline — chunked store policies into ChromaDB, retrieves semantically relevant chunks per query, and generates grounded answers via Groq LLM; built without frameworks (no LangChain), understanding every component directly.

Bangalore House Price Prediction

[GitHub](#) | [Live App](#)

Python, Scikit-learn, Flask, HTML/CSS/JS

- Engineered a `price_per_sqft` feature, applied location-wise outlier removal (mean $\pm$ std), and used One Hot Encoding to handle 200+ unique location categories.
- Selected Linear Regression via GridSearchCV achieving 81.8% accuracy (R^2); built a Flask REST API backend with an interactive HTML/CSS/JS + jQuery frontend.
- Deployed live on Render – a fully accessible end-to-end application demonstrating production ML deployment skills.

Additional Projects (GitHub): Face Mask Detector – MobileNetV2, 99.9% accuracy | MNIST Digit Classification – Dense NN, 97.49% test accuracy | NYC Taxi Fare Prediction – XGBoost, RMSE 3.9677 (60% improvement over baseline) | [Full Portfolio](#)

Education

Bachelor of Technology – Electronics & Communication Engineering

Dec 2021 – Apr 2025

Raghu Engineering College

CGPA: 8.69 / 10

M.P.C (Mathematics, Physics, Chemistry)

Jun 2019 – Mar 2021

Pandit Junior College

96.6%

Certifications

- SQL Certification – SkillUp by GeeksforGeeks | Joins, CTEs, Window Functions, Aggregations
- SQL and Relational Databases 101 (DB0101EN) – IBM Skills Network | Feb 2026
- Data Analytics Essentials – Cisco | Data workflows, statistics, analytical reasoning
- Microsoft Power BI Data Analyst (PL-300) – In Progress